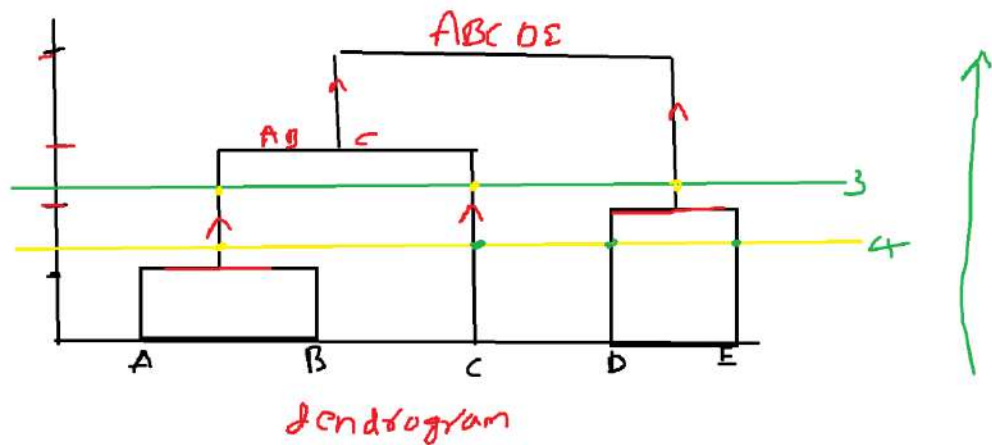


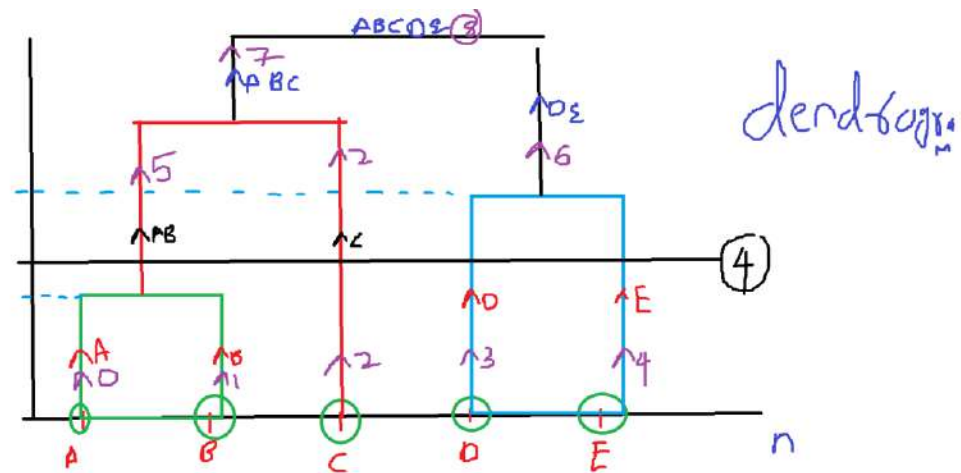
Hierarchical cluster

It is also called as **Agglomerative cluster**

It is also called as bottom to top approach

- It will assume each data point is one cluster
- Based on distances it will link the data points and form the cluster
- at top level all the data points becomes one cluster
- starting we have n data points, so we have n clusters
- at the end all the data points becomes a 1 cluster
- The clusters are varies from n to 1
- These clusters are formed based on **linkage functions**





Example: Assumption, the below table is similarity table, after calculate distances

$$S = \frac{1}{d}$$

Maximum similarity = close

similarity matrix vs Disimilarity matrix

Similarity matrix means consider the max value for cluster formation

disimilarity means consider minimum value

Similarity matrix:

the distance is less so more similar

	A	B	C	D	E
A	----	2	7	9	4
B	2	----	9	11	14
C	7	9	----	4	8
D	9	11	4	---	2
E	4	14	8	2	---

the above table is similarity tables

First calculate the distances

Then calculate the similarity = $\frac{1}{d}$

More similar means more close each other

Step – 1: which has highest similarity

$$E - B = 14$$

$A \ B \ C \ D \ E$ are 5 clusters

$B - E$: one

$A \ C \ D \ BE$

	A	C	D	BE
A	----	7(AC)	9(AD)	<fill>
C	7(CA)	----	4(CD)	<fill>
D	9(DA)	4(DC)	----	<fill>
BE	<fill>	<fill>	<fill>	----

	A	C	D	BE
A	----	7	9	2
C	7	---	4	8
D	9	4	---	2
BE	2	8	2	---

A to BE = ?

A to B : 2

A to E : 4

(2,4) how many ways pick

- 1) Max (2,4) : Single linkage : Max similarity or Min distance
- 2) Min(2,4) : Complete linkage : Min similarity or Max distance
- 3) Avg(2,4) : Average linkage

A to BE : 2

C to BE =?

C to B : 9

C to E : 8

(9,8) complete linkage= 8

C to BE =8

D to BE :

min(11,2): 2

	A	C	D	BE
A	----	7	9	2
C	7	---	4	8
D	9	4	---	2
BE	2	8	2	---

Highest pair from above table : A-D:9

BE ,AD, C

	AD	BE	C
AD	----	2	4
BE	2	---	8
C	4	8	--

AD -BE

BE we already have from above table

BE-D : 2

BE-A : 2

Complete linkage = $\min(2,2)= 2$

:2 AD-BE

AD-C

A-C:7

D-C:4

$\min(7,4)=4$

AD-C:4

	AD	BE	C
AD	----	2	4
BE	2	---	8
C	4	8	--

$H_{pair} = BE - C = 8$

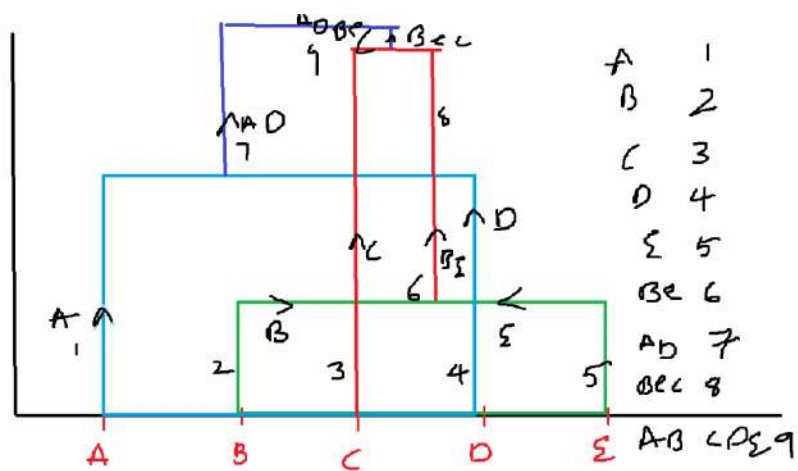
AD and BEC

Case-1: BE

Case-2: AD

Case-3: BEC

Case-4: ADBEC



Types of Linkage functions

- single linkage
 - max similarity or min distance
 - Cluster form based on less distance
 - if we want to group the data points **based on feature** then will use single
 - Irregular shapes will form
 - Affected by outliers
- Complete linkage (Generally will use)
 - min similarity or max distance
 - Cluster form based on max distance
 - if we want to group the data points based on **all the feature** then will use complete
 - well clustered
 - not affected by outliers
- Avg linkage
 - Balance between both Single and Complete

Holding hands one student with another student: Single

Not Holding hands but stay together : Complete

- Ward method : based on SSE

Ward linkage will group the cluster based on SSE or wcss

A B C D E

<i>A</i>	<i>1</i>
<i>B</i>	<i>3</i>
<i>C</i>	<i>5</i>
<i>D</i>	<i>7</i>
<i>E</i>	<i>9</i>

<i>cluster combination</i>	<i>Centriod</i>	<i>SSE</i>
<i>AB(1,3)</i>	$AB = \frac{1+3}{2} = 2$	$(1-2)^2 + (3-2)^2$
<i>AC</i>		
<i>AD</i>		
<i>AE</i>		
<i>BC</i>		
<i>BD</i>		

BE		
CD		
CE		
DE		

Minimum SSE value pair become one cluster

Assume that AD has min SSE then AD form one cluster

$\{AD\}\{B\}$ 1,7,3	$(1+7+3)/3 = 11/3 = 3.5$	$(1-3.5)^2 + (7-3.5)^2 + (3-3.5)^2$
$\{AD\}\{C\}$		
$\{AD\}\{E\}$		
$\{B\}\{C\}$		
$\{B\}\{E\}$		
$\{C\}\{BE\}$		